Mahindra University
Global Thinkers. Engaged Leaders

Mahindra University Hyderabad

École Centrale School of Engineering

Minor-II exam

Program: B. Tech. **Branch: CIE, ECM, ICS, CSE, AI, CM, NT, COM** **Year: II**

Semester: II

Subject: Numerical Methods (MA2208)

Date: 13/04/2026 **Start Time: 10:00 AM**

Time Duration: 1.5 Hours **Max. Marks: 15**

Instructions:

1) Answer all the questions.
2) All questions are self-explanatory; no clarification will be provided during the exam.
3) Use of non-programmable scientific calculator is allowed. However, sharing calculators during exams is strictly prohibited.

Question 1 (4 marks)

Consider the following data

x	-2	-1	0	1	2
$f(x)$	2.9	1.2	0.5	1.6	3.9

(a) Assume the data can be approximated by a linear function of the form:

$$p_1(x) = ax + b$$

Using the method of least squares, derive the **normal equations**.

(b) Solve the normal equations to find the values of a and b.

Question 2 (4 marks)

Discuss the convergence criteria of Gauss-Seidal method for solving the following system. Then perform two iterations with initial vector $x^{(0)} = [0, 0, 0]^T$

$$\begin{aligned} 3x_1 - 0.1x_2 - 0.2x_3 &= 7.85 \\ 0.1x_1 + 7x_2 - 0.3x_3 &= -19.3 \\ 0.3x_1 - 0.2x_2 + 10x_3 &= 71.4 \end{aligned}$$

Question 3 (4 marks)

If $A = \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix}$, then find $\|A\|_1$, $\|A\|_2$ and $\|A\|_\infty$.

Question 4 (3 marks)

Find the interpolating polynomial by using Lagrange's formula for the following data:

$$\begin{array}{rccc} x: & 0 & 2 & 5 \\ f(x): & 2 & 26 & 152 \end{array}$$

Then, estimate $f(1)$ and $f(6.5)$.
